Composed Natural Geometries for Core-Valence Diatomics: LiH and BeH$^+$ from Discrete Graph Topology*

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The natural geometry hierarchy assigns each separable quantum system a coordinate system where Coulomb singularities become boundary conditions: S^3 for atoms (Paper 7), prolate spheroids for H_2^+ (Paper 11), hyperspherical for He (Paper 13), molecule-frame hyperspherical for H_2 (Paper 15). Systems with core electrons resist a single natural geometry because core and valence electrons occupy different length scales. We introduce *composed natural geometries*—fiber bundles where each electron group inhabits its own natural coordinate system, connected by screening functions and ab initio Phillips–Kleinman pseudopotentials derived entirely from the core wavefunction. For LiH ($Z_A = 3$, $Z_B = 1$, 4 electrons): the Li$^+$ $1s^2$ core is solved via Level 3 hyperspherical (Paper 13), yielding $Z_{\text{eff}}(r)$ and pseudopotential parameters; the two valence electrons are solved via Level 4 molecule-frame hyperspherical (Paper 15) with Z_{eff} injection; the potential energy surface is scanned via a ρ-collapse adiabatic method that reduces per-R-point cost to seconds. Results: $R_{\text{eq}} = 3.21$ bohr (6.4% error vs. experiment 3.015, improved from 17% in LCAO), $\omega_e = 1471$ cm^{-1} (4.6% error vs. 1406), with zero experimental molecular data in the pseudopotential. Total wall time is $\sim$4 minutes per molecule on commodity hardware. The same formula transfers to BeH$^+$ ($Z_A = 4$) with no per-molecule tuning.

I. INTRODUCTION

The Geometric Vacuum program constructs discrete lattice Hamiltonians by identifying the *natural geometry* of each quantum system and discretizing it as a sparse graph [8–10]. Four levels of this hierarchy have been established:

Level 1: *One-center, one-electron* (H). Fock's stereographic projection [1] maps the hydrogen atom to the unit S^3, where the $1/r$ potential becomes the conformal factor. The discrete graph Laplacian reproduces hydrogenic energies to $< 0.1\%$ (Paper 7 [10]).

Level 2: *Two-center, one-electron* (H_2^+). Prolate spheroidal coordinates separate the two-center Coulomb problem exactly. Paper 11 [12] achieves 0.0002% energy error (spectral Laguerre radial solver; 0.70% with the original finite-difference lattice) and zero free parameters.

Level 3: *One-center, two-electron* (He). Hyperspherical coordinates $(R_e, \alpha, \hat{r}_1, \hat{r}_2)$ resolve the e–e cusp as a boundary condition at $\alpha = \pi/4$, $\theta_{12} = 0$. Paper 13 [14] achieves 0.05% error for He with a single adiabatic channel.

Level 4: *Two-center, two-electron* (H_2). Molecule-frame hyperspherical coordinates $(R_e, \alpha, \theta_1, \theta_2, \Phi)$ parameterized by $\rho = R/(2R_e)$ resolve all five Coulomb singularities simultaneously. Paper 15 [15] recovers 96.0% of D_e for H_2 ($l_{\max} = 6$, 61 channels, 2D variational) and 93.1% for HeH$^+$ in the *adiabatic* treatment (σ+π channels)—a non-variational over-binding figure whose consistent 2D-variational value is only $\sim$5% of D_e.

Level 4 handles H_2 and HeH$^+$ but these are *core-free* systems: all electrons participate in bonding. LiH has 2 core electrons on Li and 2 valence electrons, occupying different length scales. The LCAO approach [18] gives $R_{\text{eq}} = 2.5$ bohr (17% error vs. experiment 3.015) because the graph Laplacian's intra-atom kinetic energy is R-independent—the atomic lattice structure does not change as atoms approach [18]. Paper 15 identified this as the "locked-core boundary" problem: a single Level 4 geometry for all four electrons would require S^{11} with $\sim 10^6$ angular basis functions, computationally infeasible.

We solve this by *composing* geometries from different levels. Each electron group gets its own natural coordinate system. The inter-group coupling is handled by three typed edges: (i) screening via $Z_{\text{eff}}(r)$ from the core density, (ii) Pauli exclusion via a Phillips–Kleinman [2] pseudopotential, and (iii) nuclear repulsion via bare charges. This is the first realization of a *category* of natural geometries with morphisms preserving lattice structure.

The key results: LiH $R_{\text{eq}} = 3.21$ bohr (6.4% error), $\omega_e = 1471$ cm^{-1} (4.6% error), with zero experimental molecular data. The same code, with only (Z_A, Z_B, M_A, M_B) changed, produces a bound BeH$^+$ with physical spectroscopic constants. We do not claim production accuracy; this is a proof of concept for the composed geometry architecture.

* Paper 17 in the Geometric Vacuum series

II. THEORY: COMPOSED NATURAL GEOMETRIES

A. The composition principle

The composed graph for a core-valence diatomic is

$$G_{\text{total}} = G_{\text{nuc}} \ltimes G_{\text{core}}(R) \ltimes G_{\text{val}}(R, \text{core state})\,, \quad (1)$$

where $\ltimes$ denotes the semi-direct product (fiber bundle). Each layer inhabits a different natural geometry:

Layer 0 (nuclear):: Morse SU(2) $\times$ SO(3) from Paper 10 [11]. States $|v, J, M_J\rangle$.

Layer 1 (core):: Level 3 hyperspherical on the heavy atom at charge Z_A. States parameterized by $(R_e^{\text{core}}, \alpha_{\text{core}})$.

Layer 2 (valence):: Level 4 molecule-frame hyperspherical. States parameterized by $(R_e^{\text{val}}, \alpha, \theta_1, \theta_2, \Phi; \rho)$.

Backward compatibility between levels fails for a fundamental reason: Level 4 coordinates require two electrons (hyperradius $R_e = \sqrt{r_1^2 + r_2^2}$, hyperangle $\alpha = \arctan(r_2/r_1)$), which are undefined for a single electron. Level 4 cannot subsume Level 1. Similarly, Level 3 requires two electrons and cannot represent one. The hierarchy is a two-dimensional grid (centers $\times$ electrons), not a linear tower. Composition is necessary because no single coordinate system handles all electron groups simultaneously.

The total energy at internuclear distance R is

$$E_{\text{total}}(R) = E_{\text{core}} + V_{\text{cross}}(R) + E_{\text{elec}}(R) + V_{NN}(R)\,, \quad (2)$$

where E_{core} is the R-independent core energy from the Level 3 solve, $V_{\text{cross}}(R)$ is the core–B nuclear attraction, $E_{\text{elec}}(R)$ is the valence electronic energy from the Level 4 solve, and $V_{NN}(R) = Z_A Z_B / R$ is the bare nuclear repulsion.

The core-to-other-nucleus attraction for a $1s^2$ core is computed analytically:

$$V_{\text{cross}}(R) = -n_{\text{core}}\, Z_B \left\langle 1s_{Z_A} \left| \frac{1}{r_B} \right| 1s_{Z_A} \right\rangle\,, \quad (3)$$

where the expectation value evaluates to

$$\left\langle 1s_Z \left| \frac{1}{r_B} \right| 1s_Z \right\rangle = \frac{1}{R}\left[1 - (1 + ZR)\, e^{-2ZR}\right]. \quad (4)$$

This uses a hydrogenic $1s$ approximation for the core orbital; a more accurate treatment would integrate Z_B/r_B against the numerically computed core density from the Level 3 solve. For Li^+ ($Z = 3$), the hydrogenic approximation is accurate to $\sim$1% because the two-electron core is well-described by an effective nuclear charge $Z_{\text{eff}}^{\text{core}} \approx 2.7$.

B. Core screening function

The core is solved via the Level 3 adiabatic hyperspherical method [3] (Paper 13 [14]) at nuclear charge Z_A. The one-electron density $n(r)$ is extracted by integrating $|\psi(R_e, \alpha)|^2$ over the hyperangle and the coordinates of the second electron, using $r_1 = R_e \cos\alpha$, $r_2 = R_e \sin\alpha$. At each hyperradius R_e, the angular eigenfunction is resolved on a finite-difference grid of $n_\alpha = 200$ points, and contributions from both electrons are binned onto a radial grid.

The cumulative electron count

$$N_{\text{core}}(r) = \int_0^r n(r')\, dr' \quad (5)$$

gives the effective nuclear charge

$$Z_{\text{eff}}(r) = Z_A - N_{\text{core}}(r)\,, \quad (6)$$

with boundary conditions $Z_{\text{eff}}(r \to 0) = Z_A$ (no screening at the nucleus) and $Z_{\text{eff}}(r \to \infty) = Z_A - 2$ (full screening by two electrons). For Li^+: $Z_{\text{eff}}(0) = 3.0$, $Z_{\text{eff}}(2\text{ bohr}) \approx 1.01$.

a. Algebraic core density. The histogram-binned density can be replaced by an algebraic computation. The Level 3 angular eigenvector provides channel expansion coefficients $c_l^{(\nu)}(\alpha)$ at each hyperangle. Via the coordinate-transform identity $r_1 = R_e \cos\alpha$, the one-electron density at radius r is obtained by summing $|c_l|^2$ contributions at the hyperangle $\alpha = \arccos(r/R_e)$, weighted by the hyperradial probability $|F(R_e)|^2$. This avoids histogram binning entirely (`method='algebraic'` in `CoreScreening`).

For Li^+, the resulting $Z_{\text{eff}}(r)$ is well approximated by a closed-form expression

$$Z_{\text{eff}}(r) \approx 1 + 2(1 + ar + br^2)\, e^{-cr}, \quad (7)$$

which reproduces the ab initio curve to 0.7% accuracy. The deviations from hydrogenic screening ratios (e.g., $Z_{\text{eff}}(r)/Z_{\text{eff}}^{\text{hydro}}(r) \neq 1$) directly quantify electron correlation in the core.

C. Ab initio Phillips–Kleinman pseudopotential

The valence electrons must be orthogonal to the core. In continuous quantum chemistry, this is enforced by the Phillips–Kleinman [2] pseudopotential. We parameterize the PK barrier as

$$V_{\text{PK}}(r) = \frac{A\, e^{-Br^2}}{r^2}\,, \quad (8)$$

where A and B are derived entirely from the core wavefunction.

a. Barrier width B. The PK barrier must be localized where the core density is concentrated. We define the inverse-square-weighted core radius

$$r_{\text{eff}} = \sqrt{\frac{\int n(r)\,dr}{\int n(r)/r^2\,dr}}\,, \tag{9}$$

which weights the inner core—where $1/r^2$ is large—more heavily than the diffuse tail. Then

$$B = \frac{1}{2\,r_{\text{eff}}^2}\,. \tag{10}$$

For Li: $r_{\text{eff}} = 0.267$ bohr, $B = 7.00$ bohr^{-2}. For Be: $r_{\text{eff}} = 0.200$ bohr, $B = 12.53$ bohr^{-2}.

b. Barrier height A. The kinetic energy cost of core–valence orthogonalization is

$$A = \left|E_{\text{core}}/N_{\text{core}} - E_{\text{val}}\right| \cdot N_{\text{core}}\,, \tag{11}$$

where E_{core} is the total core energy from the hyperspherical solve, $N_{\text{core}} = 2$, and E_{val} is estimated from the first ionization energy of the neutral atom (an atomic property, not a molecular one: IE(Li) = 0.198 Ha [5], IE(Be) = 0.343 Ha [5]). For Li: $A = 6.93$ Ha·bohr^2. For Be: $A = 13.01$ Ha·bohr^2.

No experimental molecular data enters these formulas. Only atomic properties: Z, the core wavefunction (via the Level 3 solve), and the first ionization energy of the neutral atom.

c. l-dependent pseudopotential. The channel-blind PK barrier [Eq. (8)] applies the same repulsion to all angular channels, but the $1s^2$ core is pure $l = 0$. The exact PK projector $|1s\rangle\langle 1s|$ has non-zero matrix elements only for channels with $l = 0$ character. An l-dependent mode scales the per-electron PK barrier by $\delta_{l_i,0}$: valence electrons in $l > 0$ channels experience no core repulsion, consistent with the angular orthogonality of the core.

At $l_{\max} = 2$ this reduces R_{eq} error from 6.9% to 5.3% for LiH. The $l_{\max}$ divergence (Sec. V F) is reduced but not eliminated: at $l_{\max} = 3$, R_{eq} error is 14.7% (vs 16.9% channel-blind). The remaining divergence is attributed to differential angular correlation, not PK artifacts.

D. Z_{eff} injection into the Level 4 solver

The Level 4 angular Hamiltonian (Paper 15 [15]) contains nuclear attraction terms $-Z_A/r_{1A} - Z_A/r_{2A}$. We replace the constant Z_A with the function $Z_{\text{eff}}(r)$ from the core solution. In the coupled-channel expansion, the nuclear coupling matrix element between channels (l_1', l_2') and (l_1, l_2) at fixed α involves a Gaunt integral weighted by the nuclear potential. For constant Z_A, these are computed via an analytical multipole expansion. For $Z_{\text{eff}}(r)$, we add a smooth Gauss–Legendre quadrature correction for the $[Z_A - Z_{\text{eff}}(r)]/r$ difference.

Since $Z_{\text{eff}}(0) = Z_A$, the correction vanishes at $r \to 0$, eliminating the $1/r$ singularity from the quadrature. This ensures machine-precision backward compatibility when Z_{eff} is constant.

E. The ρ-collapse and fast PES scanning

Paper 15 proved that the Level 4 angular eigenvalue problem depends on (R, R_e) only through the dimensionless ratio $\rho = R/(2R_e)$. This means the angular eigenvalue curves $\varepsilon_\nu(\rho)$ can be computed once on a one-dimensional grid of ρ values and then interpolated via cubic splines.

The effective radial potential is

$$U(R_e; R) = \frac{\varepsilon_0(\rho) + 15/8}{R_e^2}\,, \quad \rho = \frac{R}{2R_e}\,, \tag{12}$$

and the electronic energy at each R is obtained by solving the one-dimensional radial Schrödinger equation

$$\left[-\tfrac{1}{2}\frac{d^2}{dR_e^2} + U(R_e; R)\right] F(R_e) = E_{\text{elec}}\,F(R_e) \tag{13}$$

via a tridiagonal eigensolver (cost: milliseconds).

The angular cache is built on a non-uniform ρ grid of $n_\rho \approx 80$ points (denser at small ρ where the eigenvalue changes rapidly), with the static Hamiltonian (kinetic, centrifugal, Liouville, e–e coupling) precomputed once and the nuclear coupling added at each ρ. Cache construction for a single R-point takes ∼2 seconds.

III. COMPUTATIONAL DETAILS

All calculations use the following parameters unless stated otherwise.

a. Core solve. The Li^+ $1s^2$ core (or Be^{2+} $1s^2$) is solved via the Level 3 adiabatic hyperspherical method with $l_{\max} = 2$, $n_\alpha = 200$, $N_R^{\text{angular}} = 200$ R-points for the adiabatic curves, and $N_R^{\text{radial}} = 2000$ grid points for the radial solve. Wall time: ∼3 seconds.

b. Angular cache. $n_\rho = 80$, $\rho \in [0.02, 5.0]$, $l_{\max} = 2$ (σ channels only; n_{ch} determined by $l_{\max}$). $n_\alpha = 100$ grid points in the angular finite-difference mesh. Cache build per R-point: ∼2 seconds.

c. PES scan. 18 R-points, $R \in [2.0, 7.0]$ bohr, with denser sampling near the minimum. Per-R radial solve: $n_{R_e} = 300$ grid points, $R_e \in [0.3, 15.0]$ bohr, tridiagonal eigensolver. Total PES scan time: ∼48 seconds.

d. Morse fit. Nonlinear least-squares fit (Levenberg–Marquardt) to the Morse potential $E(R) = E_{\min} + D_e[1 - e^{-a(R-R_{\text{eq}})}]^2$ within ±1.5 bohr of the grid minimum.

e. Reduced masses. μ(LiH) $= 1822.888 \times 7.016 \times 1.008/(7.016+1.008)$ a.u.; $\mu(BeH^+) = 1822.888 \times 9.012 \times 1.008/(9.012 + 1.008)$ a.u.

f. Total pipeline time. ∼4 minutes per molecule on a single-core commodity workstation.

g. What is not included. Core–valence exchange integrals (deferred), non-adiabatic coupling between angular channels (DBOC only), $l_{\max} > 2$ (results at $l_{\max} = 3$ cited for comparison), π channels (σ-only at $l_{\max} = 2$).

All calculations were performed with GeoVac v1.6.1 [19]. The complete pipeline (core solve, angular cache, PES scan, Morse fit, nuclear lattice) is implemented in the class `ComposedDiatomicSolver` and is reproducible with the command `ComposedDiatomicSolver.LiH_ab_initio(l_max=2).run_all()`.

IV. RESULTS

A. Ab initio pseudopotential parameters

Table I collects the PK parameters for Li and Be cores, derived entirely from the core wavefunction and atomic ionization energies.

B scales approximately as Z^2 ($7.00/12.53 \approx 9/16 = (3/4)^2$), consistent with the core radius scaling as $1/Z$.

B. LiH potential energy surface

Table II compares spectroscopic constants for LiH under different pseudopotential treatments.

Without the PK pseudopotential, the valence electrons collapse into the core ($R_{\rm eq} = 0.88$ bohr). The PK barrier is necessary, not merely helpful. The ab initio PK gives ω_e closer to experiment (4.6% error) than the fitted PK (15% error), suggesting that the $\langle 1/r^2\rangle$-weighted barrier width captures the correct PES curvature. The ab initio $R_{\rm eq}$ overshoots experiment (3.21 vs. 3.015 bohr) because the ab initio barrier height $A = 6.93$ exceeds the fitted value $A = 5.0$ by 39%, producing stronger Pauli repulsion that pushes the equilibrium outward. The barrier widths are nearly identical ($B = 7.00$ ab initio vs. 7.0 fitted), so the overshoot is controlled entirely by A. Reducing A toward 5 would recover the fitted $R_{\rm eq}$ at the cost of reintroducing a fitted parameter; we retain the ab initio value as the parameter-free prediction.

The $l_{\max}$ dependence provides a convergence indicator: $l_{\max} = 2$ gives $R_{\rm eq} = 2.97$ bohr (fitted) or 3.21 bohr (ab initio); $l_{\max} = 3$ gives $R_{\rm eq} = 3.23$ bohr (fitted). The experimental value (3.015) is bracketed by $l_{\max} = 2$ and $l_{\max} = 3$ with fitted PK, and within 6.4% at $l_{\max} = 2$ with the ab initio PK.

D_e is overestimated because the dissociation limit at $R = 7$ bohr has not fully converged to the separated-atom energy. This is a grid convergence issue, not a structural limitation.

TABLE I. Ab initio Phillips–Kleinman pseudopotential parameters. All quantities from the core wavefunction; no molecular data.

Quantity	Li (Z=3)	Be (Z=4)	Formula
$r_{\rm eff}$ (bohr)	0.267	0.200	Eq. (9)
A (Ha·bohr^2)	6.93	13.01	Eq. (11)
B (bohr^{-2})	7.00	12.53	Eq. (10)
$E_{\rm core}$ (Ha)	−7.280	−13.700	Level 3

TABLE II. LiH spectroscopic constants at $l_{\max} = 2$. "LCAO" is the atom-centered Full CI result from Ref. [18]. "No PK" disables the pseudopotential; "Ab initio PK" uses Eqs. (11)–(10); "Fitted PK" uses manually chosen $A = 5.0$, $B = 7.0$. Experimental data from Ref. [4].

Property	LCAO	No PK	Ab initio	Fitted	Expt.
$R_{\rm eq}$ (bohr)	2.5	0.88	3.21	2.97	3.015
ω_e (cm^{-1})	—	2637	1471	1617	1406
B_e (cm^{-1})	—	—	6.63	7.84	7.51
Bound?	Yes	collapse	Yes	Yes	Yes

TABLE III. BeH^+ spectroscopic constants at $l_{\max} = 2$. Reference values are indicative, from high-level ab initio calculations on light hydride ions [4].

Property	Ab initio PK	Fitted PK	Reference
$R_{\rm eq}$ (bohr)	3.25	2.82	$\sim$2.48
ω_e (cm^{-1})	2857	3088	$\sim$2222
Bound?	Yes	Yes	Yes

C. BeH^+ transferability

The same `ComposedDiatomicSolver` class, with only (Z_A, Z_B, M_A, M_B) changed, produces a bound BeH^+. Table III reports the results.

No molecule-specific code or parameter tuning is required. The PK parameters are derived from the Be^{2+} core solution using the identical formulas [Eqs. (9)–(11)].

D. Sensitivity analysis

Scanning the PK parameters independently reveals gradual, monotonic dependence:

- A scan (B fixed at 7.0): $A \in [1, 10]$ maps to an $R_{\rm eq}$ range of $\sim$1.5 bohr.
- B scan (A fixed at 6.93): $B \in [0.5, 10]$ maps to an $R_{\rm eq}$ range of $\sim$1.0 bohr.

The equilibrium is not sharply tuned to specific PK values. The physics (core screening + Pauli repulsion) determines the equilibrium, not parameter adjustment.

E. B formula comparison

Four definitions of the core radius were tested, each producing a different $B = 1/(2r_{\rm core}^2)$ and hence a different LiH equilibrium. Table IV reports the results.

TABLE IV. Effect of B formula on LiH R_{eq} at $l_{\max} = 2$. All use ab initio $A = 6.93$.

Method	r_{core} (bohr)	B (bohr^{-2})	LiH R_{eq} (bohr)	Error (%)
RMS	0.677	1.09	4.00	32.7
Peak	0.717	0.97	4.06	34.5
Median	0.500	2.00	3.75	24.4
Inv-sq (selected)	0.267	7.00	3.21	6.4

The inverse-square formula [Eq. (9)] wins because it weights the inner core—where the PK barrier must act—via the $\langle 1/r^2 \rangle$ expectation. The RMS and peak formulas are dominated by the diffuse tail and produce barriers that are too broad. The median is intermediate. This is not parameter selection—it is choosing the physically correct moment of the core density for the purpose at hand (repulsive barrier localization).

V. DISCUSSION

A. The composed geometry as a category

The composition $G_{\text{total}} = G_{\text{nuc}} \ltimes G_{\text{core}} \ltimes G_{\text{val}}$ is a fiber bundle with typed inter-layer edges. This is a mathematical structure (a category) where objects are natural geometries and morphisms are the continuous deformations ($R \to \infty$ dissociation, $Z \to Z'$ isoelectronic substitution) that transform one composed geometry into another. The LiH $\to$ BeH^+ transfer is such a morphism: the same functor (`ComposedDiatomicSolver`) maps different input parameters to different molecules.

B. What the R_{eq} improvement means

The LCAO approach gave $R_{\text{eq}} = 2.5$ bohr because the graph Laplacian's kinetic energy is R-independent: the atomic lattice structure does not change as atoms approach. The composed geometry fixes this because $Z_{\text{eff}}(r)$ encodes the core's response to the approaching H atom: at large R, the valence sees $Z_{\text{eff}} \approx 1$ (screened Li^+); at small R, the PK barrier prevents core penetration. The equilibrium is the balance between bonding stabilization (from shared valence density) and Pauli repulsion (from core orthogonalization). This balance was absent in the LCAO approach.

C. Comparison with standard quantum chemistry

The pseudopotential concept has a long history in solid-state physics [6, 7]. Standard pseudopotential methods (effective core potentials, ECPs) solve the same problem within Gaussian basis set frameworks. Our approach is structurally different: the core is solved in its natural geometry (hyperspherical), the valence in its natural geometry (molecule-frame hyperspherical), and the coupling is through the core wavefunction itself rather than through fitted ECP parameters. The analogy: ECPs are to our ab initio PK as Gaussian basis sets are to our natural geometry lattices—both solve the same physics from different starting points.

D. Limitations

1. The Gaussian/r^2 PK form [Eq. (8)] is approximate. The exact Phillips–Kleinman pseudopotential is nonlocal and energy-dependent [2]. Our Gaussian approximation captures the dominant repulsion but not its angular or energy dependence.

2. Core–valence exchange integrals are omitted for LiH (the core and valence fibers have different channel structures). For polyatomics, the full 1-RDM inter-fiber exchange reduces R_{eq} error to 11.7% (Sec. V F).

3. Only σ channels at $l_{\max} = 2$. Paper 15 showed that π channels add $\sim$7 percentage points for H_2; similar improvement is expected for LiH at $l_{\max} \geq 3$.

4. The dissociation limit is not converged at $R = 7$ bohr. D_e values are upper bounds.

5. Only two-core plus two-valence systems. Extension to more electrons requires composing additional Level 3/4 subproblems (e.g., lone pairs as Level 3, additional bonds as Level 4).

A natural question is whether PK can be eliminated in a quantum simulation context, where antisymmetry is enforced by the creation/annihilation algebra rather than by explicit orthogonalization. A scoping investigation (Track CB, v2.0.37)[21] tested replacing PK with explicit cross-block two-electron integrals for LiH: the result is negative. Cross-block ERIs add core–valence electron repulsion, but the composed orbital basis (different Z_{eff} per block) means no cross-center nuclear attraction is present in h_1. Without the balancing one-body attraction, the Hamiltonian is energetically unbalanced: 4-electron FCI gives 29% error (worst of three configurations tested), with the Pauli count increasing 2.56× and 1-norm 2.30×. PK thus remains the correct coupling mechanism for the composed architecture in *classical* solvers. In quantum settings, Paper 19 [20] demonstrates that the missing one-body cross-center integrals $\langle \psi^A_{nlm} | V_B | \psi^A_{n'l'm'} \rangle$ can be computed analytically via incomplete gamma functions (the multipole expansion terminates exactly at $L_{\max} = 2l_{\max}$ by Gaunt selection rules). The resulting "balanced coupled" Hamiltonian—with all four interaction types and no PK—produces the only bound LiH in 4-electron FCI (1.8% energy error vs PK's 15.0%) and achieves 0.20% energy error at

$n_{\max} = 3$. For BeH_2, PK overcorrects: 27.4% error in 4-electron FCI vs 10.7% balanced. The balanced 1-norm is lower than composed for BeH_2 (323.4 vs 374.9 Ha, both identity-included). See Paper 19 for the full resource census and convergence analysis.

E. Architectural note: two solvers, two drift signatures

The composed-geometry framework comprises two distinct PES solvers with different operating regimes and drift signatures, both of which appear under the umbrella name "composed Hamiltonian":

1. **Level-4 PK-composed solver** (in `geovac/composed_diatomic.py`): the level-4 hyperspherical PES solver of Sec. IV that produces the headline LiH 5.3% $R_{\rm eq}$ error at $l_{\max} = 2$. Its $R_{\rm eq}$ drift slope is +0.15–0.22 bohr per unit $l_{\max}$, structural to PK overcounting at higher angular channels.

2. **Balanced-coupled solver** (in `geovac/balanced_coupled.py`, with n_e-projected FCI diagonalization in `geovac/coupled_composition.py`): the qubit-encoded Hamiltonian builder used for resource benchmarking (Paper 14, Paper 20). The balanced architecture replaces PK entirely with cross-block ERIs plus cross-center V_{ne} via Shibuya–Wulfman multipole expansion (Paper 19). Its $R_{\rm eq}$ drift slope is +0.054 bohr per unit $n_{\max}$, a 3× improvement over the PK-composed solver.

These are different solvers in different code paths with different working bases ($l_{\max}$-truncated hyperspherical channels vs. $n_{\max}$-truncated single-center hydrogenic orbitals on each block). Phase C "W1c-aware" multi-focal-composition closures (Paper 19, Sec. "The W1c-residual orthogonality wall") for $Z \geq 11$ frozen-core hydrides apply to the balanced-coupled path only; the PK-composed path is unaffected by them. Empirically, for first-row LiH at $n_{\max} = 2$, the W1c screening machinery is bit-exactly equivalent to the bare path (verified to 5×10^{-14} Ha across a 5-point PES grid) because no [Ne]-style frozen core is present. The chemistry-solver upgrade already shipped at v2.0.39 (the balanced-coupled architecture itself); subsequent multi-focal-composition closures address second-row and heavier systems.

The post-Phase-C engineering closures for the balanced-coupled second-row chemistry-solver wall (W1c-residual orthogonality) have landed two clean architectural modules without recovering binding yet: `geovac/phillips_kleinman_cross_center.py` (W1b PK barrier, Sprint 2026-05-08) and `geovac/screened_valence_basis.py` (Track 3, Sprint 2026-05-09; replaces hydrogenic Z_orb=1 valence h_1 diagonal with the actual screened Schrödinger eigenvalue, e.g. Na 3s = −0.170 Ha). Both modules are bit-exact backward-compatible with the existing path (LiH 2,726-Pauli regression preserved at machine precision) and are correct as architectural infrastructure. The negative result is on the binding outcome: NaH PES remains monotonically descending under all 8 combinations of {bare, W1c, PK, SV}. See Paper 19 Sec. "The W1c-residual orthogonality wall" for the per-track diagnostics. The Track-3 diagnostic localizes the genuine residual to wavefunction shape (not eigenvalue), with the actual Na 3s screened wavefunction having mean radius $\langle r \rangle = 4.47$ bohr versus the hydrogenic Z=1 1s mean radius of 1.50 bohr. Wavefunction-shape substitution in cross-V_{ne} and within-block ERIs is the next named engineering closure (sprint-scale via physical-n hydrogenic relabeling, or beyond sprint scale via strict-Schmidt orthogonalization).

a. Update post-Sprint α arc (2026-05-23): a third layer beneath the wavefunction-shape correction. The modular propinquity reframing (Sprint M-Y, see Paper 19 Sec. "The W1c-residual orthogonality wall") sharpened the wavefunction-shape correction into a two-axis L/R bimodule decomposition with predicted right-action dominance $d_R/d_L \gg 1$. The α-Diagnostic confirmed this empirically ($d_R/d_L = 3.23$ at canonical probe width); the α-Multi-zeta sprint built clean architecture for the named target (mixed-Slater-n physical Na 3s/3p fits with overlap > 0.999999, wired into `geovac/multi_zeta_orbitals.py` Z=11 entry, `shibuya_wulfman.py` multi-zeta dispatch, and `balanced_coupled.py` `multi_zeta_basis` kwarg); the α-PES three-step gate exposed a substantive new structural finding before the production wiring was committed.

The W1c-residual wall has a third structural layer beneath the two prior engineering closures targeted. At NaH $n_{\max} = 2$, the 2-electron FCI ground state is structurally H-side localized: the lowest 5 eigenvalues of h_1 are entirely H-localized (dragged deep by the un-screened $Z = 11$ cross-V_{ne} attraction on H), and the Na 3s eigenvalue sits above them at $i = 5$ in the eigenspectrum, unoccupied in the ground state. Multi-zeta substitution of the Na 3s basis shifts the Step-1 algebraic kernel differential by −0.135 Ha at $R_{\rm eq}$ (correctly, the operator-level architecture works), but produces a bit-zero FCI eigenvalue shift ($|\Delta E| < 4 \times 10^{-13}$ Ha across all tested R) because the FCI ground state doesn't occupy the Na valence space. Layer-3 is invisible to any on-site h_1 or on-site basis correction; it lives in the FCI variational state itself.

Three named follow-on directions surface from the α arc, ranked in Paper 19 Sec. "The W1c-residual orthogonality wall": (i) NaH $n_{\max} = 3$ test bundled with W1c+multi-zeta unified architecture (sprint-scale; falsifies/confirms Layer 3 as a basis-size artifact); (ii) cross-V_{ne} kernel-shape substitution (sprint-scale, parallel diagnostic of Track 3's original named target); (iii) removal of the screened-path multi-zeta `NotImplementedError` (bundled with item (i)). The architectural addition from α-Multi-zeta + α-PES is retained per the architectural-cleanliness rule. Synthesis memo: `debug/sprint_modular_alpha_arc_synthesis_memo.md`.

b. Update post-Sprint F1 arc (2026-05-23): structural success / energetic failure split at $n_{\max} = 3$. The three named follow-ons above were executed in a single-day F1 arc (items (i) and (iii) bundled; item (ii) deferred as the natural F2 next target). The unified W1c $\times$ multi-zeta architecture (item (iii) done) ran at NaH $n_{\max} = 3$ ($Q = 56$) with three falsifiable predictions frozen before the test: P1 (internal PES minimum at $R_{\rm eq} \in [3.0, 4.5]$ bohr), P2 ($D_e \in [0.0375, 0.150]$ Ha within $2\times$ experimental), P3 (multi-zeta differential $\in [0.02, 0.30]$ Ha at $R_{\rm eq}$). **Verdict: CLEAN NEGATIVE** per the pre-specified gate. P1 fails on the exact falsification criterion ($R_{\min} = 2.0$ bohr, smallest tested); P2 fails (descent depth 0.71 Ha, $\sim 10\times$ experimental — spurious-binding signature); P3 passes (mz differential 0.21 Ha at $R = 2.0$, consistent with $n_{\max} = 2$'s 0.21 Ha at the same R — endomorphism content preserved across basis sizes).

The substantive new structural finding for the composed-architecture context after the full F1$\to$F2$\to$F3 arc is the architectural-extension ladder that closes W1d at the FCI level while opening W1e. F1 max_n=3 (2026-05-23) showed that at $n_{\max} = 3$ under combined W1c+mz the dominant natural orbital IS a true bonding combination ($-0.698 \cdot \phi_{\rm Na,\,3s} - 0.687 \cdot \phi_{\rm H,\,1s}$ with 50/50 amplitude^2 split), confirming basis enlargement closes the orbital-mixing-flexibility gap. F2 (2026-05-23) then ran a cross-V_{ne} kernel-shape substitution diagnostic before committing to a production extension: the multipole kernel at default $L_{\max} = 4$ is bit-faithful to converged 3D quadrature within $\sim 10^{-5}$ Ha (six to ten orders of magnitude below wall depth), **ruling out kernel-shape substitution as the W1c-residual closure mechanism**. F2's substantive new content was the diagnosis that the framework's h1 in `composed_qubit.build_composed_hamiltonian` is strictly block-diagonal — the off-diagonal h1 slot connecting orbitals on different centers is **architecturally absent** (W1d). F3 (2026-05-23) closed W1d at the FCI level: new production module `geovac/cross_block_h1.py` computes the missing cross-block h1 elements via 2D axial Gauss-Legendre quadrature; bit-exact backward compat preserved; `build_composed_hamiltonian` extended with `cross_block_h1` kwarg. At NaH max_n=2 with the full W1c+mz+xblockh1 stack the FCI naturals transform from $[1.0000, 1.0000]$ (W1c+mz alone, two separated singly-occupied orbitals) to $[\mathbf{1.9991}, \mathbf{0.0007}]$ (one doubly-occupied bonding orbital, 50/50 Na/H mix) — a four-orders-of-magnitude advance in the dominant natural-orbital occupation signature. **W1d is CLOSED at the FCI level.** But the F3 PES remains monotonically descending with well depth $D_e^{\rm F3} = +4.37$ Ha $= 58\times$ experimental: cross-block h1 introduces bonding without supplying opposing Pauli repulsion against the [Ne] frozen core. **New sub-layer W1e (inner-region overattraction)** is the natural F4 target via a bonding-orbital Phillips-Kleinman extension (extends existing `geovac/phillips_kleinman_cross_center.py` which currently operates on partner-side valence diagonal). The two-bucket M-Z partition (cross-shift / module endomorphism) stands with architectural-absence as a structurally distinct third sub-category WITHIN the partition (F2 sharpening, F3 confirmation). Comprehensive full-arc synthesis memo: `debug/sprint_w1c_full_arc_synthesis_memo.md`; sprint memos `debug/sprint_f2_cross_vne_kernel_memo.md` (W1d named) and `debug/sprint_f3_cross_block_h1_memo.md` (W1d closed, W1e named). F1-maturity Sprint β-neg synthesis at `debug/sprint_beta_neg_synthesis_memo.md` (superseded for framing by the full-arc memo).

c. Refinement of W1e after F4–F6 (2026-05-23 β-update). Three diagnostic sprints in immediate succession (F4, F5, F6) tested the candidate W1e closure mechanisms F3 had named, and refined W1e from "missing Pauli repulsion" to a multi-mechanism deep-correlation wall. **F4** (bonding-orbital Phillips–Kleinman extension): RULED OUT at Step 1 — bonding-core overlap $|S| = 17.5\%$ on Na 2s is substantial but the PK barrier is only $+0.194$ Ha ($22\times$ smaller than wall depth); rank-1 PK saturation gives at most 43% closure ceiling. **W1e is not a single-particle Pauli-orthogonality wall**; it is a multi-determinant FCI correlation effect. **F5** (explicit-core Hartree treatment in FCI): RULED OUT at Step 1 — cross-block 2-body Coulomb $J - K$ between bonding orbital and Na [Ne] core $= +1.12$ Ha (right SIGN, repulsive) but only 25.7% of the 4.37 Ha wall, predicting residual $D_e^{\rm F5} \approx +3.25$ Ha $= 43\times$ experimental. **W1e is not addressable by mean-field Hartree-level core-bonding interaction. F6** (max_n=4 valence basis enlargement, full F3 stack): PARTIAL_IMPROVEMENT, only 10.2% closure at PES level (2-point gate suggests 26.1% but this is artifactual; the inner-region collapse continues from $R = 3.5$ down to $R = 2.0$ with another 0.696 Ha descent the 2-pt gate misses). **W1e is not addressable by basis enlargement at max_n=4. Net refinement:** W1e decomposes into three structurally INDEPENDENT sub-sub-mechanisms each with 10–25% closure ceiling: basis-truncation ($\sim$ 10% PES, F6) + Hartree-pressure ($\sim$ 25% 2-pt, F5) + deep-correlation-residual ($\sim$ 65–90% of wall, requires an architectural lift beyond sprint scale: Schmidt orthogonalization at basis-construction level, or full-correlation [Ne] cores). The W1e closure remains OPEN with a clean three-sub-sub-mechanism characterization. Methodological side finding: 2-point gates at fixed reference geometry overestimate closure fractions by $\sim 2.5\times$ for monotonically-descending PES in this regime; future sprints should use PES well depth at $R_{\min}$ as the load-bearing metric. Sources: `debug/sprint_f4_bonding_pk_memo.md`, `debug/sprint_f5_explicit_core_memo.md`, `debug/sprint_f6_maxn4_nah_memo.md`, `debug/sprint_beta_update_f4f6_memo.md`; refined Paper 19 § "Refinement of the W1e mechanism after F4–F6"; Paper 34 § "W1e three-sub-sub-mechanism refinement after F4–F6"; Paper 32 §VIII Sprint M-Z addendum F4–F6 extension.

d. Closure of the W1e sprint-scale candidate space (Schmidt + core correlation Day-1 diagnostics, 2026-05-23). The two named beyond-sprint-scale candidates from the F4–F6 refinement (Schmidt orthogonalization at basis-construction level; fully correlated [Ne] cores) were tested by Day-1 diagnostics and both ruled out. Schmidt orthogonalization of H 1s against Na [Ne] core at NaH $R = 3.5$ bohr produces $\sum_c |S_c|^2 = 1.00 \times 10^{-3}$ (0.1% projection mass) and a total diagonal-element differential $\Delta_{\rm total} = +8.6$ mHa — $\sim 12\times$ below the GO threshold; the H atom is structurally far outside the [Ne] core's significant amplitude. Literature order-of-magnitude estimates for the binding-relevant [Ne] core-correlation contribution converge to ~ 0.001–0.003 Ha, $10^3\times$ below the 4.37 Ha wall depth. *Five independent sprint-scale W1e closure mechanisms now ruled out* (F4 single-particle PK, F5 mean-field Hartree, F6 basis enlargement, Schmidt, core correlation). The chemistry-side architectural-extension ladder reaches a natural pause at sprint scale; the W1e wall is now characterized as the sixth instance of the framework's multi-focal-composition wall pattern (first in chemistry, joining five physics observables documented in CLAUDE.md §1.7: Sprint H1 Yukawa, LS-8a $Z_2/\delta m$, HF-3 recoil, HF-4 Zemach, HF-5 multi-loop a_e). The structural-skeleton scope statement extends from physics-only to physics+chemistry. Sources: `debug/sprint_w1e_schmidt_diagnostic_memo.md`, `debug/sprint_w1e_core_correlation_estimate_memo.md`; refined Paper 19 § "Closure of the W1e sprint-scale candidate space"; Paper 34 § "Cross-domain instance of the multi-focal-composition wall"; Paper 32 §VIII Sprint M-Z addendum Schmidt + core correlation extension.

F. Outlook: the composition pattern for larger molecules

The composed geometry generalizes:

- **BeH_2:** 1 core (Be $1s^2$) + 2 bond pairs (Level 4 on each Be–H) + inter-bond coupling.
- **H_2O:** 1 core (O $1s^2$) + 2 bond pairs (Level 4 on each O–H) + 2 lone pairs (Level 3 on O) + inter-pair coupling.
- **General:** partition electrons into cores (Level 3 per atom) + bond pairs (Level 4 per bond) + lone pairs (Level 3 on atoms). The molecule is a *hypergraph* where each hypernode is a natural-geometry subproblem and each hyperedge is a typed coupling.

We tested LiH at $l_{\max} = 2$–7 with l-dependent PK (applying the pseudopotential barrier only to $l = 0$ channels, where the $1s$ core resides), expecting the $l_{\max}$ convergence pattern from Paper 15 (H_2) to reduce $R_{\rm eq}$ error. The result is the opposite: $R_{\rm eq}$ diverges monotonically from experiment.

$l_{\max}$	$N_{\rm ch}$	$R_{\rm eq}$ (bohr)	Error	$\Delta R_{\rm eq}$
2	5	2.80	7.0%	—
3	8	3.02	0.0%	+0.21
4	13	3.42	13.5%	+0.41
5	18	3.47	15.2%	+0.05
6	25	3.56	18.1%	+0.09
7	32	3.64	20.7%	+0.08

a. Mechanism: differential angular correlation. The $l_{\max}$ divergence arises from *differential angular correlation*: higher-l channels provide angular correlation energy that preferentially stabilizes large-R (diffuse) geometries over small-R (compact) geometries, shifting the PES minimum outward. A diagnostic decomposition confirms this is entirely nuclear in origin. Removing all electron–electron off-diagonal coupling while retaining the full nuclear coupling gives a ground-state channel weight $w(0{,}0) = 0.243$ and 8 channels for 90% of the norm—massive spreading driven by the asymmetric nuclear potential. Conversely, retaining only the e–e coupling (diagonal nuclear) gives $w(0{,}0) = 0.9998$—the electron–electron interaction produces no channel mixing. The spreading is 100% nuclear.

b. Eigenvalue convergence at fixed geometry. At fixed internuclear geometry (R, R_e), the angular eigenvalue μ_0 converges exponentially with $l_{\max}$ at rate $\beta \approx 0.42$ per unit $l_{\max}$ (exponential fit $R^2 = 0.997$). The divergence of $R_{\rm eq}$ is therefore not a failure of eigenvalue convergence but a differential effect: the converged eigenvalue improves faster at large R than at small R, tilting the PES. The overall drift rate is +0.168 bohr per unit $l_{\max}$ (linear fit, $R^2 = 0.89$).

c. R-dependent pseudopotential scaling. The self-consistent $l = 0$ content of the valence wavefunction decreases with R: at $l_{\max} = 4$, the $l = 0$ fraction drops from 0.85 at $R = 2.0$ bohr to 0.33 at $R = 6.5$ bohr. To compensate, we scale the PK barrier strength with R:

$$w_{\rm PK}(R) = \delta_{l_i,0} \times \min\!\big(\text{cap},\ R/R_{\rm ref}\big)\,, \tag{14}$$

where $R_{\rm ref}$ is a reference distance and the cap prevents unbounded growth. At $R < R_{\rm ref}$, the PK operates at standard l-dependent strength; at $R > R_{\rm ref}$, it strengthens to counteract the loss of effective core exclusion.

For LiH with $R_{\rm ref} = 3.015$ bohr and cap = 1.5:

Mode	$l_{\max}$	$R_{\rm eq}$ (bohr)	Error
l-dependent	4	3.42	13.5%
R-dependent	4	3.08	2.0%
Self-consistent	4	3.19	5.9%
l-dependent	7	3.64	20.7%
R-dependent	7	3.61	19.7%

The R-dependent scaling reduces $R_{\rm eq}$ error from 13.5% to 2.0% at $l_{\max} = 4$, but the advantage erodes at higher $l_{\max}$: the asymptotic drift rates are $+0.168$ (l-dependent) and $+0.160$ (R-dependent) bohr per unit $l_{\max}$ over the range $l_{\max} = 2$–7. Self-consistent PK (iteratively adjusting the barrier strength based on the eigenstate's $l = 0$ content) achieves intermediate drift reduction (50%). The scaling delays the onset of divergence but does not eliminate the asymptotic drift, which reflects the fundamental limitation of the composed geometry's static core–valence partition.

d. PK-free diagnostic (negative result). A fundamental test of the PK pseudopotential's role is to disable it entirely (`pk_mode='none'`). Without PK, the valence electrons are not excluded from the core region, and the resulting PES is *monotonically attractive* at all $l_{\max}$: no equilibrium geometry exists. The PK barrier is therefore essential for *creating* the equilibrium, not merely refining it. The balance between bonding stabilization and Pauli repulsion (Sec. V B) requires an explicit core-exclusion mechanism; screening via $Z_{\rm eff}(r)$ alone is insufficient.

This diagnostic shows only that PK is required for an equilibrium to exist at all: without PK there is no minimum to diverge. It does *not* localize the $l_{\max}$ divergence to the adiabatic approximation; the variational analysis below shows the 2D solver produces the *same* drift, so the divergence is intrinsic to the composed-geometry architecture, not the adiabatic approximation or the PK form. The operating point remains $l_{\max} = 2$.

e. Ab initio $R_{\rm ref}$ derivation (negative result). The reference distance $R_{\rm ref}$ in Eq. (14) was originally set to $R_{\rm eq}^{\rm expt}$, introducing a non-ab-initio parameter. Three atomic core-derived candidates were tested as ab initio replacements:

- Core orbital radius $\langle r \rangle_{1s} = 0.50$ bohr
- PK Gaussian width $1/\sqrt{B} = 0.27$ bohr
- Screening length (radius where $Z_{\rm eff} = Z_A - 1$): 0.82 bohr

All three are too small to affect the PES: an $R_{\rm ref}$ sweep confirms that $R_{\rm ref} \leq 2.0$ bohr has zero effect on $R_{\rm eq}$, while only $R_{\rm ref} \sim 3.0$ bohr ($\approx R_{\rm eq}$ itself) produces reasonable $R_{\rm eq}$ error. The fundamental problem is a *scale mismatch*: $R_{\rm ref}$ must be molecular-scale ($\sim R_{\rm eq}$) to modulate $w_{\rm PK}(R)$ in the bonding region, but $R_{\rm eq}$ is a molecular property not derivable from atomic core data alone. Deriving $R_{\rm ref}$ from core properties creates circular reasoning: the only atomic length scales (0.27–0.82 bohr) are too small by a factor of $\sim$4–11.

f. Root cause: adiabatic approximation with asymmetric charges. Subsequent diagnostics (v2.0.6) have traced the $l_{\max}$ divergence to the adiabatic multichannel approximation interacting with PK-induced angular symmetry breaking. A bare HeH^+ Level 4 calculation—with no $Z_{\rm eff}$, no Phillips–Kleinman pseudopotential, and no composition—exhibits the same qualitative drift: $+0.262$ bohr per $l_{\max}$. In composed LiH, where $Z_A = Z_B = 1$ (symmetric), the PK barrier on atom A breaks the $A \leftrightarrow B$ symmetry, producing an effective heteronuclear system with drift $+0.303$ bohr per $l_{\max}$. The mechanism is that asymmetric nuclear coupling populates odd-l channels forbidden in homonuclear systems; H_2 places 94% of its channel weight in the (0,0) channel, while HeH^+ places only 68%, with 20% in odd-l channels that preferentially stabilize large-R configurations.

Three alternative mitigation strategies were tested and found insufficient. (i) An algebraic PK projector (rank-1 $|\psi_{\rm core}\rangle\langle\psi_{\rm core}|$) produces drift 5.6× worse than Gaussian PK ($+1.697$ vs $+0.303$ bohr/$l_{\max}$), because the valence wavefunction avoids the core state in function space while still penetrating the core region in coordinate space. (ii) Enhanced $Z_{\rm eff}$ with a repulsive dip (PK-free) requires extreme parameters ($Z_{\rm eff}(0) = -38$) due to $1/r^2$ vs $1/r$ shape mismatch, and the odd-l channel population rises to 30.7%—worse than bare HeH^+. (iii) Single-channel DBOC correction of $+0.035$ Ha is 97% cancelled by off-diagonal P-matrix coupling in the complete-basis limit, overcorrecting by 23×.

g. Complete solver–PK survey. All four solver × PK combinations have now been tested. Table V summarizes $R_{\rm eq}$ error and $l_{\max}$ drift rate for each configuration, including updated fine-grid results (16 R-points) from Track BQ.

TABLE V. LiH $R_{\rm eq}$ error (%) vs. experiment (3.015 bohr) for all solver × PK combinations. "Drift" is the average $R_{\rm eq}$ increase per unit $l_{\max}$ (bohr/$l_{\max}$). Adiab. = adiabatic single-channel; 2D = variational (R_e, α) solver. Ch-blind = channel-blind PK; l-dep = l-dependent PK ($\delta_{l,0}$ projection). The "2D + ch-blind (coarse)" row uses the original 9-point R-grid; the "2D + ch-blind (fine)" row uses a 16-point R-grid that resolves the shallow PES minimum more accurately.

Configuration	l=2	l=3	l=4	Drift
Adiab. + ch-blind	6.4	16.9	32.8	+0.40
Adiab. + l-dep	5.3	— — —	— — —	—
2D + ch-blind (coarse)	6.1	9.5	— — —	+0.10
2D + ch-blind (fine)	5.5	10.4	20.4	+0.15–0.30
2D + l-dep	6.1	9.5	16.1	+0.15

h. Updated root cause (v2.0.32). Fine-grid analysis (16 R-points, Track BQ) revealed that the earlier coarse-grid survey overestimated the adiabatic contribution to the drift. On the fine grid, the 2D variational solver produces drift of $+0.15$–0.30 bohr/$l_{\max}$ (Table V), comparable to the adiabatic solver's $+0.40$ bohr/$l_{\max}$. Single-point energies at fixed geometry ($R = 3.015$ bohr) also diverge with $l_{\max}$: $E = -8.184$, -8.204, -8.236 Ha at $l_{\max} = 2$, 3, 4 respectively (exact: -8.071 Ha), violating the variational bound and confirming PK overcounting as the structural source. CBS extrapolation gives $R_{\rm eq} \to 3.734$ bohr (23.8% error). The $l_{\max}$ divergence is therefore intrinsic to the composed-geometry PK architecture, not to the adiabatic approximation. The optimal operating point is $l_{\max} = 2$.

Three conclusions emerge from the full survey:

1. The 2D solver does *not* substantially reduce drift relative to the adiabatic solver. The earlier coarse-grid result ($+0.10$ bohr/$l_{\max}$) was an artifact of insufficient R-grid resolution; the fine-grid drift ($+0.15$–0.30 bohr/$l_{\max}$) is comparable to the adiabatic drift ($+0.40$ bohr/$l_{\max}$).

2. The l-dependent PK has *zero effect* on the 2D solver: $R_{\rm eq}$ is identical to channel-blind PK at $l_{\max} = 2$ (6.1%) and $l_{\max} = 3$ (9.5%). The PK projection and the variational (R_e, α) treatment are apparently orthogonal corrections.

3. The drift is *not* a residual effect—it is the *entire* $l_{\max}$ divergence mechanism, intrinsic to the PK/composed-geometry architecture. Higher-l channels interact with the PK barrier in ways that systematically overcount correlation, as confirmed by the variational-bound violation at fixed geometry.

The $l_{\max}$ divergence is intrinsic to the composed-geometry approximation itself—the core/valence separation with $Z_{\rm eff}$ injection and PK exclusion—not to the adiabatic approximation or the PK form. All solver and PK combinations are now exhausted.

i. Composed-geometry accuracy ceiling. The operating point for production LiH calculations is $l_{\max} = 2$, where all four configurations give 5–6% $R_{\rm eq}$ error. The accuracy cost of the composition is quantified by the full N-electron scoping study: a 4-electron mol-frame hyperspherical solver on S^{11} (SO(12) angular basis) would have $\sim$13,000 angular channels at $l_{\max} = 2$ (vs. Level 4's $\sim$90), a 144$\times$ ratio. S_4 singlet symmetry reduction yields a $\sim$6$\times$ compression, but the resulting $\sim$3.3 days per PES point at $l_{\max} = 2$ is intractable for production use. This 144$\times$ angular compression is what the composed geometry purchases; the $\sim$5% $R_{\rm eq}$ error at $l_{\max} = 2$ is the price.

The PK pseudopotential is not an improvable approximation: it is the irreducible cost of factorizing the full antisymmetric wavefunction into composed coordinates (Paper 18, composition exchange constant). Three graph-native alternatives to PK were investigated and all failed: (i) node exclusion has no coordinate correspondence between the core S^3 and valence Level 4 graphs; (ii) fiber bundle Berry connection is intra-group, with the exchange operator P_{ij} mixing coordinate systems; (iii) spectral orthogonality in function space does not imply coordinate-space exclusion. The master obstruction is that antisymmetry requires a shared coordinate system, and composition factorizes $\Psi_{\rm total}$ into incompatible coordinates.

j. Cusp correction in the composed pipeline. The Schwartz cusp correction (Paper 15, Sec. VI.J) has been wired into the composed-geometry pipeline as a per-fiber post-processing step. For LiH at $l_{\max} = 2$, the cusp correction is 0.23 mHa at $R_{\rm eq}$, which is sub-dominant to both the PK-induced error and the adiabatic approximation error. $R_{\rm eq}$ is unchanged by the cusp correction. The correction is available via the `cusp_correction=True` parameter in `ComposedDiatomicSolver`.

Initial BeH_2 results validate the qualitative PES shape (bound state, repulsive wall, correct dissociation limit) but reveal a 32% $R_{\rm eq}$ error with the block-diagonal approximation. The core-valence composition validated for LiH is insufficient for valence-valence coupling between bond pairs sharing the Be nucleus: $Z_{\rm eff}$ partitioning and classical inter-bond repulsion both fail to improve accuracy. Orbital-level inter-bond coupling (exchange and orthogonalization between bond-pair wavefunctions) is the bottleneck for quantitative polyatomic PES. The Pauli term scaling ($Q^{2.5}$, Paper 14) is unaffected by this limitation, as it depends on the block structure, not the inter-block coupling.

1. *Inter-fiber coupling investigation*

A systematic three-phase investigation of inter-fiber coupling confirms that the channel overlap between bond-pair fibers is the key quantity for polyatomic accuracy.

a. Monopole Coulomb coupling (negative result). The Slater F^0 integral between the two bond-pair densities, transformed to Be-centered coordinates, gives a monopole inter-fiber energy of $\sim$1.6 Ha that is nearly R-independent near equilibrium. Adding this scalar repulsion worsens $R_{\rm eq}$ from 32% to 62% error—the same failure mode as classical inter-bond repulsion and $Z_{\rm eff}$ partitioning.

b. Inter-fiber channel overlap $S_{\rm avg}(R)$. The parity-weighted overlap between ground adiabatic channels of the two fibers under π-rotation is

$$S(\rho) \;=\; \sum_{l_1,l_2} (-1)^{l_1+l_2} \left|c^0_{l_1 l_2}(\rho)\right|^2, \qquad (15)$$

where $c^0_{l_1 l_2}$ are the channel expansion coefficients extracted from the angular eigenvector at each $\rho = R/(2R_e)$, and $S_{\rm avg}(R)$ is the $|F(R_e)|^2$-weighted average over the hyperradius. This is a purely algebraic quantity—no spatial integration is required.

$S_{\rm avg}$ has strong R-dependence: it drops from 0.47 at $R = 2.0$ bohr to 0.24 at $R = 6.0$ bohr, with slope $dS/dR = -0.16$ at $R_{\rm eq}$. The physical origin is a shift in channel composition: the (0,0) σ_s channel weight drops from 61% at short R to 20% at large R as odd-parity channels (0,1) and (1,0) grow. Even-parity channels contribute positively to S; odd-parity channels contribute negatively. The net effect is a monotonic decrease in S as the wavefunction redistributes angular momentum.

c. Exchange coupling via $S \cdot F^0$ (partial success). Combining the overlap and the monopole integral as $E_{\rm exch}(R) = -S_{\rm avg}(R) \cdot F^0(R)$ reduces $R_{\rm eq}$ error from 31% to 20%. The ab initio coupling captures $\sim$40% of the correction identified by a single-parameter fit ($K_{\rm eff} \approx 1.6$ vs $K_{\rm fit} = 4.08$ Ha). The remaining $\sim$60% is recovered by including the off-diagonal 1-RDM elements (see "Full one-particle density matrix exchange" below).

The inter-fiber channel overlap $S(R)$—a purely algebraic quantity computed from angular eigenvector coefficients and parity phases—captures the R-dependent coupling between bond-pair fibers. This validates the fiber bundle framework for polyatomic molecules: each bond pair keeps its own natural geometry, and the coupling between fibers is mediated by quantum number overlap at the shared center.

d. Channel-decomposed F^0 and direct exchange. The monopole integral F^0 can be decomposed by angular channel, giving channel-specific direct Coulomb integrals $F^0_{\rm ch}$. A direct exchange formula $E_{\rm direct} = -\sum_{\rm ch}(-1)^{l_1+l_2}F^0_{\rm ch}$ avoids the $S \cdot F^0$ factorization and gives similar accuracy: $R_{\rm eq}$ error of 20.0% (vs 19.2% for the $S \cdot F^0$ path).

e. Higher Slater multipoles (negative result). Including $k = 1$ (dipole) and $k = 2$ (quadrupole) multipoles in the inter-fiber coupling does not recover the missing exchange. The $k = 1$ term is *repulsive* and worsens $R_{\rm eq}$; $k = 2$ is negligibly small. The total multipole correction $k = 0 + 1 + 2$ gives 20.5% $R_{\rm eq}$ error, slightly worse than monopole alone (20.0%). This confirms that the missing exchange comes from off-diagonal 1-RDM terms, not higher angular multipoles.

f. Full one-particle density matrix exchange. The channel-resolved one-particle reduced density matrix $\gamma^A_{l_1,l_1'}$ is obtained by tracing electron 2 from the Level 4 ground-state eigenvector:

$$\gamma^A_{l_1,l_1'} = \sum_{l_2}\sum_{j} v_{(l_1,l_2),j}\, v_{(l_1',l_2),j}, \tag{16}$$

where the sum runs over all channels sharing the same l_2 and over the α grid, with the result $|F(R_e)|^2$-weighted over the hyperradius. The off-diagonal elements $\gamma_{l_1\neq l_1'}$ are 68–121% of the diagonal norm (growing with R as channel mixing increases).

The full inter-fiber exchange between identical fibers A and B in the channel representation is

$$K_{AB}(R) = -\sum_{l_1,l_1'}\left[\gamma^A_{l_1,l_1'}\right]^2 (-1)^{l_1+l_1'}\, F^0_{l_1,l_1'}(R), \tag{17}$$

where $F^0_{l_1,l_1'}$ is the l_1-marginal Slater monopole integral computed from Be-centered densities aggregated over l_2, and the parity factor arises from $\gamma^B_{l_1',l_1} = (-1)^{l_1+l_1'}\gamma^A_{l_1',l_1}$ under the π-rotation relating the two fibers. The diagonal limit $l_1 = l_1'$ recovers a quantity proportional to the existing $S \cdot F^0$ exchange; the off-diagonal terms capture inter-channel exchange.

Method	$R_{\rm eq}$ (bohr)	Error vs 2.507
Block-diagonal (no coupling)	3.28	31%
Monopole F^0 (Phase 1)	4.06	62%
Exchange $S \cdot F^0$	3.01	20%
Full 1-RDM exchange [Eq. (17)]	2.80	11.7%
Fitted model ($-K \cdot S$)	2.81	12%

The full 1-RDM exchange matches the single-parameter fitted model with zero free parameters. The off-diagonal terms are subtractive ($\sim$10% of the diagonal magnitude at $R_{\rm eq}$), moderating the exchange and providing the correct R-dependence. This confirms that the off-diagonal γ elements—not higher multipoles or kinetic corrections—carry the missing exchange physics.

g. Kinetic orthogonalization (negative result). Löwdin symmetric orthogonalization of overlapping fiber wavefunctions raises the kinetic energy by

$$\Delta T(R) = \frac{1}{2}\sum_{l_1,l_1'}\left|S^{AB}_{l_1,l_1'}\right|^2 \frac{T_{l_1}+T_{l_1'}}{2}, \tag{18}$$

where $S^{AB}_{l_1,l_1'} = (-1)^{l_1'}\gamma^A_{l_1,l_1'}$ and $T_{l_1} = l_1(l_1+1)/(2\langle R_e^2\rangle)$ is the centrifugal kinetic energy. The correction is +0.03 Ha and nearly R-independent (it peaks at $R \sim 4.0$ where $l \geq 1$ channel growth counteracts overlap decay). Adding ΔT to the full exchange does not shift $R_{\rm eq}$. The residual 11.7% error is attributed to basis truncation ($l_{\rm max} = 2$) and the adiabatic approximation.

h. Transferability to core–valence exchange (LiH). The full 1-RDM exchange is specific to the bond–bond (Level 4 + Level 4) topology of polyatomics like BeH_2. In LiH, the two fibers are the Level 3 core (atom-centered hyperspherical) and the Level 4 valence (molecule-frame hyperspherical)—fundamentally different channel structures that cannot share γ matrices. The PK pseudopotential already encodes core–valence orthogonality, so explicit exchange between these fibers is expected to be partially redundant. Extending the 1-RDM exchange to heterogeneous fiber pairs would require a cross-level overlap formalism mapping Level 3 angular channels onto Level 4 channels, which is deferred to future work.

i. Path forward. The full 1-RDM exchange closes the gap between the diagonal $S \cdot F^0$ approximation and the fitted model. The remaining 11.7% $R_{\rm eq}$ error points to three directions: (a) $l_{\rm max}$ convergence with full exchange (current results use $l_{\rm max} = 2$), (b) non-adiabatic corrections (DBOC), and (c) cusp corrections at the coalescence point. Kinetic orthogonalization has been tested and found negligible. Self-consistent Hartree iteration is deprioritized, as the kinetic correction is uniform and SCF is unlikely to provide differential R-dependence.

A fourth direction—transcorrelated (TC) integrals in the qubit pipeline—was explored (Tracks BX-3/TC-V, v2.0.35) but found to be a dead end. The TC transformation $H_{\rm TC} = e^{-J} H\, e^{J}$ with $J = \frac{1}{2}r_{12}$ replaces the Coulomb ERI with a smooth gradient operator within each composed block, formally removing the $1/r_{12}$ cusp. An initial benchmark appeared to show TC converging to a 3.3–3.6% plateau for He where the standard encoding diverged (5.3% $\to$ 8.2%), but that divergence was an artifact of diagonalizing the full qubit (Fock) space including wrong-particle-number sectors; under correct particle-number-projected full CI the standard pipeline converges to $\sim$2.0% and does not diverge, so TC's $\sim$3.4% plateau is in fact *worse* than the untransformed result.

The cusp is an energy-evaluation problem, not a wavefunction problem (cf. Paper 13), and TC is not pursued further. Cross-block perturbative MP2 corrections were also investigated but found negligible ($<$ 0.5 mHa for LiH).

j. H_2O: first triatomic PES. The composed geometry extends to H_2O as a five-block decomposition: O $1s^2$ core (Level 3, $Z = 8$), two O–H bond pairs (Level 4 with charge-center origin, $Z_{\text{eff}} = 6$, $Z_B = 1$), and two lone pairs (Level 3, $Z_{\text{eff}} = 6$). The architecture introduces no free parameters. Inter-fiber coupling at the H–O–H bond angle $\theta = 104.5°$ generalizes the linear-molecule rotation $(-1)^{l_1+l_2}$ to Wigner d-matrix phases $P_{l_1}(\cos\theta) \cdot P_{l_2}(\cos\theta)$, recovering the linear formula at $\theta = \pi$. Six coupling pairs arise: one bond–bond, four bond–lone, and one lone–lone.

The uncoupled PES with charge-center origin gives $R_{\text{eq}} = 1.459$ bohr (19.4% error vs 1.809 experimental), with the correct qualitative shape: repulsive wall, well-defined minimum, and smooth dissociation. (This 19.4% figure, improved from the earlier 26% / 1.34 bohr recorded when this result was first written, reflects general solver evolution since then—*not* the inter-fiber PK-consistency fix, which the *uncoupled* PES reported here does not invoke.) The bond–bond coupling ($\sim$0.5 Ha) is consistent with BeH_2 and barely affects R_{eq}. However, the bond–lone and lone–lone couplings are unphysical: the $S \cdot F^0$ factorization produces -28 Ha (bond–lone, 4 pairs) and -15 Ha (lone–lone) at the experimental geometry—magnitudes that exceed the total electronic energy. The Slater integrals scale as $\sim Z_{\text{eff}}$, and at $Z_{\text{eff}} = 6$ the compact lone pair densities drive F^0 far beyond the regime validated on BeH_2 ($Z_{\text{eff}} = 2$). Lone pair coupling is disabled for production use; only bond–bond coupling is validated.

The dominant bottleneck is the Level 4 angular basis at high charge asymmetry ($Z_A/Z_B = 6$:1 for O–H bonds). The bare bond pair R_{eq} error scales from 0.1% at 1:1 (H_2) to 18% at 2:1 (BeH) to 56% at 6:1 with midpoint origin; the charge-center origin $z_0 = R(Z_A - Z_B)/[2(Z_A + Z_B)]$ reduces the 6:1 error to 17%, but does not fully resolve the basis limitation at $l_{\max} = 2$ with 9 channels. Improved angular basis convergence for asymmetric bonds—via non-uniform grids or adaptive channel selection—is the path to quantitative H_2O accuracy.

VI. CONCLUSION

Composed natural geometries solve the core-valence problem without a single high-dimensional coordinate system. The key results:

1. LiH R_{eq} improves from 17% error (LCAO) to 5.3% (composed, ab initio PK, l-dependent) with zero molecular fitting.

2. The ab initio PK pseudopotential is derived entirely from the core wavefunction: A from the core–valence energy gap [Eq. (11)], B from the $\langle 1/r^2 \rangle$-weighted core radius [Eq. (9)].

3. The architecture transfers to BeH^+ with no per-molecule tuning.

4. The composition pattern—core graphs + valence graphs + typed coupling edges—provides a concrete framework for extending discrete graph quantum chemistry to general molecules.

5. For polyatomics (BeH_2), the full 1-RDM inter-fiber exchange [Eq. (17)] reduces R_{eq} error from 20% (diagonal $S \cdot F^0$) to 11.7%, matching a single-parameter fitted model (12%) with zero free parameters.

6. All four solver $\times$ PK combinations have been exhausted (Table V). The 2D variational solver does *not* substantially reduce the drift (the earlier "4$\times$" reduction was a coarse-grid artifact, retracted in v2.0.32; the fine-grid 2D drift is comparable to the adiabatic drift); l-dependent PK has zero additional effect on the 2D solver. A residual +0.15–0.30 bohr/$l_{\max}$ drift is intrinsic to the composed-geometry approximation. Without PK, no equilibrium exists in the classical composed solver. (Paper 19 [20] shows that the balanced coupled Hamiltonian—which replaces PK with explicit cross-block ERIs and cross-center V_{ne}—produces equilibrium in 4-electron FCI, and is the *only* coupling scheme that does so.) Ab initio R_{ref} derivation fails (scale mismatch). The PK pseudopotential is classified as a *composition exchange constant* (Paper 18): the irreducible cost of factorizing the N-electron wavefunction into composed coordinates. The operating point is $l_{\max} = 2$ (5–6% R_{eq} error), where the 144$\times$ angular compression vs. a full 4-electron solver justifies the approximation.

The natural geometry hierarchy is now:

Level	System	Geometry
1	H	S^3 (Fock)
2	H_2^+	Prolate spheroid
3	He	Hyperspherical
4	H_2	Mol-frame hypersp.
4N	LiH	Full mol-frame hypersp. (SO(12)) R_{eq}: 63.5
5	LiH	Composed (Level 3+4)
5	BeH^+	Composed (Level 3+4)
5	BeH_2	Composed (Level 3+4+4)
5b	LiH/BeH_2/H_2O	Balanced coupled (Paper 19)

VII. FULL N-ELECTRON VALIDATION

The composed geometry (Level 5) approximates an exact N-electron treatment by factorizing the wavefunction into core and valence groups with separate coordinate systems. The PK pseudopotential is the cost of this factorization (Sec. VI). To validate the approximation and quantify its cost, we constructed the exact 4-electron mol-frame hyperspherical solver for LiH—Level 4N—and compared it directly against the composed architecture.

A. Level 4N: the exact N-electron generalization

Level 4N extends the Level 4 mol-frame hyperspherical coordinates from 2 to N electrons. For LiH ($N = 4$), the 4 electrons live on S^{11} (the unit sphere in $\mathbb{R}^{12}$), with angular symmetry group SO(12) replacing Level 4's SO(6). The S_4 permutation group enforces antisymmetry, replacing the gerade constraint of the 2-electron case. Specifically, the $[2, 2]$ Young diagram of S_4 projects onto the singlet sector.

The Jacobi tree coordinates for 4 electrons produce 11 hyperangles parameterized by the hyperradius ρ. The angular eigenvalue problem is solved at each ρ by diagonalizing the angular Hamiltonian restricted to the S_4 singlet ($[2, 2]$) sector.

B. Results by $l_{\max}$

a. $l_{\max} = 0$. The S_4 $[2, 2]$ representation is *empty* at $l_{\max} = 0$: no antisymmetric 4-electron states can be constructed from s-waves alone. No PES can be computed.

b. $l_{\max} = 1$. Only 2 channels survive S_4 projection. The PES is monotonically attractive with no equilibrium—insufficient angular differentiation to produce a bonding minimum.

c. $l_{\max} = 2$. With 12 S_4 $[2, 2]$ channels and angular dimension 2592 ($n_{\text{grid}} = 6$), the first equilibrium appears:

- $R_{\text{eq}} \approx 1.0$–1.1 bohr (64% error vs. experimental 3.015 bohr)
- $E_{\min} \approx -8.42$ to -8.46 Ha (overbinding ∼5% vs. exact -8.07 Ha)
- $D_e \approx 0.44$–0.49 Ha (exact: 0.092 Ha)
- Wall time ∼69 s per PES point

This is a *structural* result: molecular equilibrium exists without PK. The 64% R_{eq} error and 5× overbinding indicate that $l_{\max} = 2$ is far from convergence for the 4-electron system—higher angular momentum channels are needed for quantitative accuracy. At $l_{\max} = 3$ (40 S_4 channels, angular dimension 2560), R_{eq} remains at ∼1.0 bohr (67% error), unchanged from $l_{\max} = 2$. The well deepens dramatically ($D_e = 1.19$ Ha vs 0.43 Ha at $l_{\max} = 2$) but the equilibrium position does not shift outward toward experiment. This confirms slow convergence: the raw SO(12) basis adds angular correlation that lowers E uniformly but cannot differentiate core from valence efficiently enough to produce the correct bond length at tractable $l_{\max}$. Channel decomposition at $l_{\max} = 2$ reveals that the dominant channels are $(0, 0, 1, 1)$ and $(1, 1, 0, 0)$ at 42% total weight—one electron pair in s-wave (core) and one in p-wave (valence)—confirming that the composed geometry's core-valence separation reflects the exact wavefunction's structure.

C. The 144× compression justification

At $l_{\max} = 2$, the angular dimension comparison is:

Solver	Angular dim	Wall time/point
Level 4N (full 4e)	2,592	∼69 s
Level 5 (composed)	∼18	∼0.1 s

The composed geometry achieves a 144× angular compression by factorizing the 4-electron problem into a 2-electron core (Level 3 on S^3) and a 2-electron valence (Level 4 on S^5), coupled via Z_{eff} screening and the PK pseudopotential.

This compression comes at the cost of the PK approximation: the composed solver achieves R_{eq} within 5–6% at $l_{\max} = 2$, while the full solver gives 64% error at the same $l_{\max}$. However, the composed solver's advantage grows with $l_{\max}$ (wall time scales as $l_{\max}^2$ for composed vs. $l_{\max}^{11}$ for the full solver). At $l_{\max} = 3$, the full solver requires ∼2500 spectral basis functions (Track AK), while the composed solver remains in the tens.

PK is classified as a *composition exchange constant* (Paper 18): the irreducible cost of factorizing the N-electron wavefunction into composed coordinates. The full N-electron solver validates this classification by demonstrating that exact S_4 antisymmetry produces equilibrium without PK, confirming that PK approximates—rather than creates—the physics of Pauli exclusion in the composed architecture.

a. Independent validation via nested hyperspherical encoding (v2.5.0). Track DF tested the alternative to composed factorization: placing all 4 electrons in a single S^{11} Hilbert space with H-set coupled hyperspherical harmonics. Three molecular variants were tested for LiH: (i) single-center (all orbitals at $Z = 3$): correct dissociation energy (4.7% error) but $R_{\text{eq}} = 2.0$ bohr (33.7% error)—orbitals too compact on the hydrogen side; (ii) two-center (charge-center origin): catastrophic energy degradation (48.2% error)—the truncated orbital basis cannot represent the $1s^2$ core density off-center; (iii) heterogeneous (per-pair Z_{eff}): R_{eq} improved to 17.1% error, but Löwdin orthogonalization destroyed Gaunt sparsity (4,743 vs. 288 Pauli terms, a 16.5× inflation) and introduced BSSE-like overbinding. The composed architecture's block-diagonal factorization is therefore not merely a computational convenience—it is the structurally optimal decomposition for multi-scale electronic systems. Each block maintains its own orthonormal basis at the appropriate length scale, avoiding the cross-scale orthogonalization penalties that afflict unified-basis approaches.

D. Non-adiabatic coupling diagnosis

The adiabatic solver results at $l_{\max} = 2$–3 (Sec. VIII.B) show worsening D_e overcounting with $l_{\max}$ (0.49 $\to$ 1.19 Ha vs. exact 0.092 Ha). A coupled-channel radial solver was constructed to test whether non-adiabatic P/Q coupling corrects this overcounting, following the Level 3 pattern (Paper 13).

The coupled-channel approach *fails* at Level 4N for numerical reasons that are structurally different from Level 3:

1. **Near-degenerate channels.** The S_4-projected eigenvalue gaps are 0.01–0.05 in the bonding region ($R_e \sim 1.5$–3.0 bohr), compared to $O(1)$ gaps at Level 3. Since $P \sim V_{\text{coupling}}/\Delta E$, the P-matrix norms are 10–30$\times$ larger (peak $\|P\|_F = 3.3$, DBOC $= 5.4$ Ha vs. Level 3's ~ 0.15 Ha).

2. **No analytic P-matrix.** Unlike Level 3, where $H(R) = H_0 + R \cdot V_C$ is a linear pencil with R-independent $dH/dR = V_C$, the 4-electron angular Hamiltonian depends nonlinearly on R_e through $\rho = R/(2R_e)$. Finite-difference P-matrices on a sparse grid are noisy and ARPACK-seed-dependent (0.478 Ha variation between runs).

3. **Truncated Q-matrix.** With 3–12 channels from a total of $\sim$2592 (S_4-projected), the closure approximation $Q \approx PP$ captures $< 0.1\%$ of external channel contributions.

Adding more channels makes results *worse*: the 6-channel solver overshoots the adiabatic energy by 1.09 Ha at $R_e = 1.0$ (wrong direction).

This is a *numerical* failure—the adiabatic representation is the wrong basis for near-degenerate channels—not a physics limitation. Non-adiabatic coupling is important; the coupled-channel formalism simply cannot compute it reliably when channel gaps are $O(0.01)$.

The resolution is a 2D variational solver that bypasses the adiabatic representation entirely. Feasibility analysis (Track AP) shows that a partial 2D treatment of (R_e, α_2)—the inter-pair hyperangle controlling the core/valence partition—produces a 2,250-dimensional tensor product at $l_{\max} = 2$, comparable in cost to the current adiabatic solver. The structural findings of Sec. VIII.B (equilibrium exists without PK, core-valence separation confirmed in the exact wavefunction) are unaffected by the solver choice and stand independently.

a. Note added v2.0.24. The 2D variational solver has been implemented and run (Track AR, v2.0.23). The 38,400-dimensional tensor product at $l_{\max} = 2$ gives $E_{\min} = -7.79$ Ha (variational bound respected, above exact -8.07 Ha) with $D_e = -0.19$ Ha (unbound). This confirms the adiabatic $D_e = +0.49$ Ha as an artifact and establishes that the $l_{\max} = 2$ S_4 $[2, 2]$ angular basis is genuinely insufficient for LiH binding. All three radial solvers (adiabatic, coupled-channel, 2D variational) are now exhausted. The composed geometry's 144$\times$ angular compression is essential: Level 5 at the same $l_{\max} = 2$ achieves 5.3% R_{eq} error vs 63.5%. Track AS (v2.0.23) further confirms that the composed qubit encoding is categorically sparser than full N-electron encoding (838 Pauli terms at $Q = 30$ vs 3,288 at $Q = 10$, 20$\times$ lower 1-norm), validating the composed architecture for both classical and quantum computation. The classical solver investigation is now complete (v2.0.24); the quantum simulation cost advantage is the primary computational result.

b. Note added v2.5.0. The necessity of the composed factorization has been independently validated by a 6-sprint investigation (Track DF) that tested nested hyperspherical alternatives on S^{11} with H-set coupled angular bases. All three molecular variants failed to simultaneously achieve correct equilibrium geometry and Gaunt-level sparsity, confirming that the block-diagonal architecture is structurally required for multi-scale molecular systems. See Paper 14 for the novel $6j$ recoupling sparsity result that emerged from this investigation.

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[21] Sprint/track designations are anchors into the project's internal chronicle (`CHANGELOG.md` in the repository).